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Dr Johannes Wachs

Corvinus University of Budapest, Hungary

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Re: Research Assistant / PhD Student positions, Geography of Code group

Dear Dr Wachs,

Developers leave traces on GitHub and LinkedIn much as the public leaves them on X and Reddit, and much of my career has gone into building the systems that collect and read the second kind. Your group asks how the digital economy, and now AI, changes who does skilled work and where. Your advertisement asks for computational methods applied to social questions, real experience in coding, modelling and data wrangling, and an interest in agentic AI. I bring all three from social media research and industry: an MSc dissertation that became a methods paper on measuring opinion across platforms, social media data systems for global brands at M+C Saatchi Fluency in London, and AI agents that I now build into that work. I would like to turn these skills on a change I am living through as a developer myself.

Your work. The paper I keep returning to is *Who is using AI to code?* with Daniotti, Feng and Neffke (*Science*, 2026). It estimates that AI wrote about 29% of US developers' Python functions by the end of 2024 and raised their output by about 3.6%, and that the gains went to experienced developers, not to the early-career developers who used AI more. That matches what I see: AI assistance helps me because years of writing such code by hand let me judge its output quickly. The paper is open about its limits, open-source Python on GitHub in six countries, and calls for widening the analysis to countries at different income levels. Your paper with del Rio-Chanona and Laurentsyeva (*PNAS Nexus*, 2024) adds a caution I take seriously: Stack Overflow activity fell by about 25% after ChatGPT's release, relative to sites where it was less available, so the public traces we study are changed by the technology under study. Together they point to the problem I most want to work on: measuring a change in work when that change also alters what the data can see.

Computational methods for social questions. My MSc dissertation in Data Analytics at the University of Portsmouth (Distinction), supervised by Dr Ella Haig, compared public reaction to UK economic policy across about 279,000 posts from X, Reddit, YouTube and Quora. Its weakness taught me most: the classifier was checked against labels seeded by a lexicon, so the check was partly circular. That became a paper with Dr Haig, *Is the Platform Part of the Measurement?*, which treats each platform as a survey mode. In a simulation I built, pooling platform scores at face value gives the wrong sign of daily aggregate sentiment on roughly a quarter of days. We are posting it to arXiv and plan an ICWSM submission.

Coding, modelling, data wrangling and databases. At Dubizzle Labs in Lahore (2022 to 2024), the company behind the Dubizzle, Bayut, Zameen and OLX marketplaces, I designed its first web-scraping architecture on AWS, collecting listings and online discussion across 15 markets in English, Arabic, Urdu and Hindi, and built the data models and machine learning behind its competitor-intelligence dashboards. I was promoted twice and led a team of three engineers. At Fix.com, a Canadian replacement-parts business, I built OCR models that parse parts catalogues from PDFs and models that backfill missing parts data. I work in Python and SQL, with PySpark, PostgreSQL, Redshift and Athena for large data, and I have designed layered data warehouses.

Agentic and generative AI. At Fluency I designed a self-healing agent that diagnoses and repairs our forum crawlers when a site changes, and a builder agent that creates new crawlers, while I decide what goes into production. As co-founder and CTO of Artemis AI, a social-media intelligence start-up, I lead a team of five building a platform whose labelling runs from our own NLP models to DeepSeek and then Claude when confidence is low, with image and video understanding, ahead of a beta launch on 15 December 2026. I read widely to make it accurate, from Grimmer, Roberts and Stewart's *Text as Data* to work on validating LLM annotation, and we are designing evaluation around human-labelled data so that every model is checked against people before it reaches users.

Why social media, and my connection to your question. Social media is the work I care most about, because it is where I learned what online numbers can and cannot say about people. GitHub and LinkedIn are platforms too, and their traces carry the problem my paper studies: what is visible depends on each platform's rules. I also have a direct connection to the developers your *Science* paper could not reach. I started as a computer science student in

Pakistan taking web-scraping orders on Fiverr, completed more than 300, and helped other students and junior colleagues into freelance and remote careers; I still pass on what I learned through web-scraping-guide.com, a free public guide to web data collection. I know how this work is found and done, how much of it stays in private client repositories, and the people who do it.

A question I could contribute to. How do AI coding agents change the division of labour in software work, and for whom? Agents now leave signatures in public traces: co-author trailers in commits, bot accounts that open pull requests, and the code patterns your classifier detects. Following developers' commits, collaboration networks and, through LinkedIn, their careers before and after they start working with agents, one could ask whether agent use changes who leads a project and the uneven workloads that your paper with Betti, Gallo and Battiston (*Nature Communications*, 2025) links to success, and whether developers in lower-income countries, including those who enter global software work through freelance platforms, see the gains your *Science* paper finds. I would check how much of a measured change comes from visibility rules rather than behaviour (a trailer on by default for one tool and absent for another, or work moving into private repositories) against repositories where true agent use is known, starting with my own.

Why a PhD, and why now. I want to turn what I learned in the MSc, at Fluency and in building Artemis AI into research rigorous enough to trust. Industry taught me to build systems that collect and label online data at scale; it did not teach me to show causally what those data say about people and their work. Your group would give me that training, and I would carry the results back into how social media and data products are built and evaluated.

What I do not yet bring. I have no peer-reviewed publication yet. My statistics are applied: hypothesis tests, the bootstrap, forecasting and simulation. I have not been trained in econometrics and causal inference, such as the difference-in-differences design of your Stack Overflow paper, and I have used network tools without formal training in network science. I would close these gaps through the doctoral school's courses and by applying the methods on the group's projects from my first month as a research assistant.

Practicalities. I am available full time from 1 December 2026 and would relocate to Budapest. My referees are Dr Ella Haig, Associate Professor in AI, School of Computing, University of Portsmouth, my dissertation supervisor and co-author (Ella.Haig@port.ac.uk), and Dr Fahad Ahmad, Senior Lecturer, Centre for Cybercrime and Economic Crime, University of Portsmouth, who taught and assessed me on the MSc (Fahad.Ahmad@port.ac.uk). Thank you for considering my application.

Yours sincerely,

Asad Ikram